

ECO 344 - Applied Econometrics I

Chapter 4

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- 2 Sampling Distribution of OLS Estimators
- 3 Testing Hypotheses About a Single Parameter
- 4 Confidence Intervals
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Inference: Introduction

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- So, the expected value of our estimator is equal to the population parameter and not the individual value of estimator we obtained from the sample.
- However, we can test some statements about our population parameters using the estimators of our sample. This is called Inference.

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- Before we dive into inference, we need to know the sampling distribution of the OLS estimators.

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- Therefore, we will make an assumption about the distribution of OLS estimators.

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- Under these assumptions, OLS is the minimum variance unbiased estimator, the best among all unbiased estimators and not only among those linear in y.

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 - ② Also, if the factors affect y in a non-additive way, for example multiplicatively, the central limit theorem argument no longer applies.
- we will try to address these concerns in chapter 5.

Testing Hypotheses About a Single Parameter

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 - Reject the Alternate Hypothesis

Under the CLM assumptions:

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1} = t_{df}$$

This is known as t-test statistics/t-ratio/t-calculated. It measures how many standard errors $\hat{\beta}_j$ lies away from your hypothesized value.

- **Rejection Rule:**

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- The critical value is the number of standard deviation that your hypothesized value can be away given any probability or significance level. It will be different given any significance level.

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- **Type II Error** - Unable to reject the Null hypothesis, When it is actually false.
 - We can control for the type II error through number of observation and the variance of OLS estimators.
 - The higher the number of observations or the lower variance of OLS estimators, the lower the probability of Type II - Error.

Two-Tailed Test of Significance



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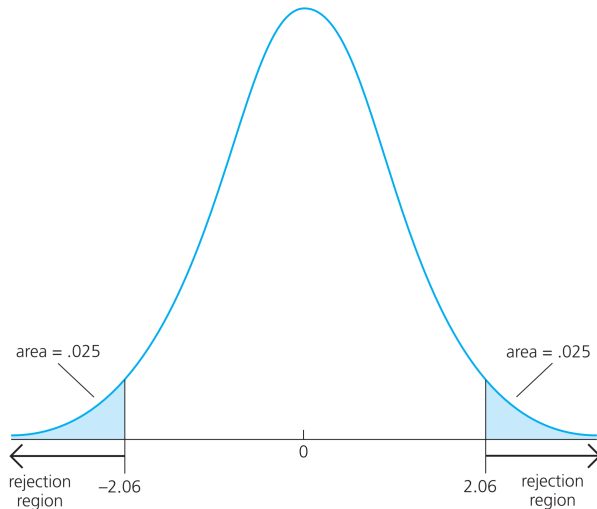
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- If we reject the null hypothesis, then we would say that we reject the null hypothesis of insignificance and conclude that x_j is statistically significant or x_j has a statistically significant effect on y .

Two-Tailed Test of Significance

Figure 4.4 5% rejection rule for the alternative $H_1: \beta_j \neq 0$ with 25 *df*.



Two-Tailed Test of Significance

Example:

EXAMPLE 4.3

Determinants of College GPA

We use the data in GPA1 to estimate a model explaining college GPA (*colGPA*), with the average number of lectures missed per week (*skipped*) as an additional explanatory variable. The estimated model is

$$\widehat{colGPA} = 1.39 + .412 \text{ } hsGPA + .015 \text{ } ACT - .083 \text{ } skipped$$
$$(.33) \quad (.094) \quad (.011) \quad (.026)$$
$$n = 141, R^2 = .234.$$

$$t_{141-3-1=137} = \frac{0.412}{0.094} = 4.38$$

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- The critical value at $df = 137$ and $\alpha = 0.05$ & 0.01 is approximately 1.96 & 2.58 respectively, using the large sample values (exact values 1.98 & 2.61). (we will see the tabulated value at 0.025 and 0.005)

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- As $t_{137} = 4.38 > 1.96 \& 2.58$, we reject the null hypothesis of insignificance and conclude that High School GPA has a statistically significant effect on the college GPA.

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- If we reject the null hypothesis, we conclude that β_j is statistically different from a at the chosen significance level.

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Determinants of College GPA

We use the data in GPA1 to estimate a model explaining college GPA (*colGPA*), with the average number of lectures missed per week (*skipped*) as an additional explanatory variable. The estimated model is

$$\widehat{colGPA} = 1.39 + .412 \text{ } hsGPA + .015 \text{ } ACT - .083 \text{ } skipped$$

(.33) (.094) (.011) (.026)

$$n = 141, R^2 = .234.$$

Test if $\beta_j = 0.2$



$$t_{141-3-1=137} = \frac{0.412 - 0.2}{0.094} = 2.26$$

- The critical value at $df = 137$ and $\alpha = 0.05$ & 0.01 is approximately 1.96 & 2.58 respectively, using the large sample values (exact values 1.98 & 2.61). (we will see the tabulated value at 0.025 and 0.005)

Two-Tailed Test of a Certain Value

Example:

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- As $t_{137} = 2.26 > 1.96$, we reject $H_0 : \beta_j = 0.2$ at the 5% level and conclude that the effect of High School GPA differs from 0.2. However, as $2.26 < 2.58$, we fail to reject it at the 1% level.

One-Tailed Test



$$H_0 : \beta_j \leq a$$

$$H_A : \beta_j > a$$



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- The t-test statistics would be;

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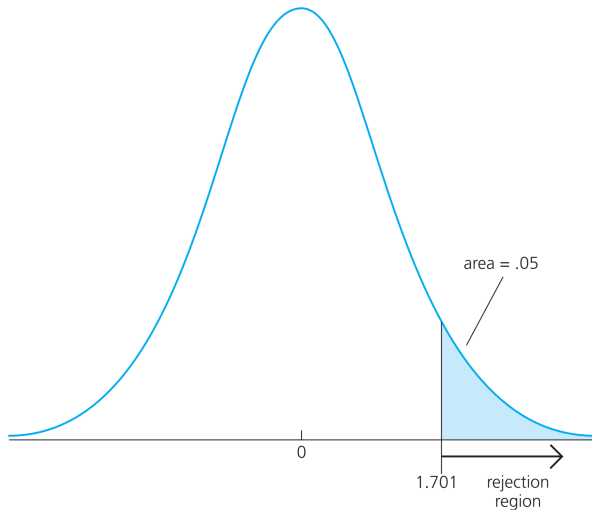
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- The critical value of t would come from the table at given significance level. (the tabulated value is taken at α)
- we will reject the null hypothesis at

$$t_{n-k-1} > c$$

One Tailed Test

Figure 4.2 5% rejection rule for the alternative $H_1: \beta_j > 0$ with 28 *df*.



Computing P-Values for t-Test

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Computing P-Values for t-Test

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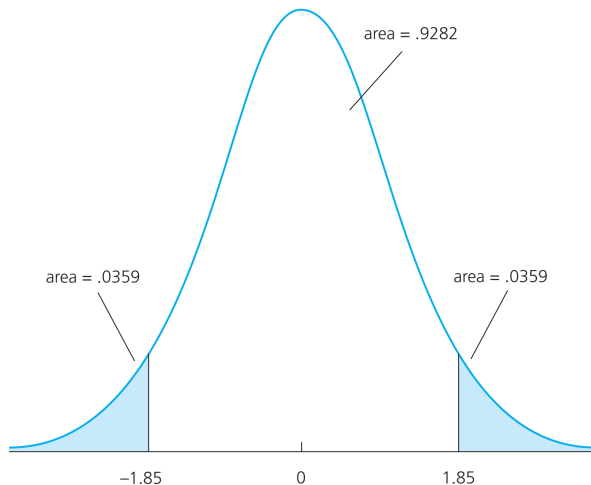
- Formally, the p-value is the probability that a t distributed random variable, under H_0 , is larger in absolute value than the observed test statistic.

$$P(|T| > |t|)$$

Where T is a random variable following the t_{n-k-1} distribution under H_0 & t is the observed test statistic.

Computing P-Values for t-Test

Figure 4.6 Obtaining the p -value against a two-sided alternative, when $t = 1.85$ and $df = 40$.



Example

```
. regress price mpg rep78 headroom
```

Source	SS	df	MS	Number of obs	=	69
Model	148497605	3	49499201.8	F(3, 65)	=	7.51
Residual	428299354	65	6589220.82	Prob > F	=	0.0002
				R-squared	=	0.2575
				Adj R-squared	=	0.2232
Total	576796959	68	8482308.22	Root MSE	=	2566.9

price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
mpg	-289.3462	62.53921	-4.63	0.000	-414.2456	-164.4467
rep78	670.8971	343.5213	1.95	0.055	-15.16242	1356.957
headroom	-300.0293	398.0516	-0.75	0.454	-1094.993	494.9346
_cons	10921.33	2153.003	5.07	0.000	6621.487	15221.17

Different Kinds of Significances

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- 3 **Practical Significance** - Wooldridge uses this term interchangeably with economic significance. Is a statistically significant but tiny effect more meaningful than a large but imprecisely estimated one?
- 4 DO NOT forget that you are economist and not statistician. Always look beyond just technical stuff.

Confidence Intervals

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- Under the CLM assumptions, we can easily construct confidence interval for OLS parameters.

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- It is important to note that confidence intervals are as good as the validity of the CLM assumptions.

Example:

$$\widehat{\log(rd)} = -4.38 + 1.084 \log(sales) + .0217 \textit{profmarg}$$
$$(.47) \quad (.060) \quad (.0128)$$
$$n = 32, R^2 = .918.$$

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$$0.9664 \leq \beta_{sales} \leq 1.2016$$

Testing Hypotheses about a Single Linear Combination of Parameters

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$$\log(wage) = \beta_0 + \beta_1 jc + \beta_2 univ + \beta_3 exper + \mu$$

where *jc* is number of years attending a two-year college.

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- It can also be written as:

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- Think about any one-tail test hypothesis involving single linear combination of parameter.

Testing Hypotheses about a Multiple Linear Combination of Parameters

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- Let's look at the following model: $\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + \beta_5 \text{rbisyr} + \mu$ where bavg , hrunsyr , rbisyr are their performance indicators in the game.

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- When we want to test the effect of one performance indicator, we will do the t-testing. However, if we want to test that does performance affect the salary?, we have to do the joint testing.

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- To do this test, we will estimate two models.
 - 1 UnRestricted Model - Same as the last slide (Original Model)
 - 2 Restricted Model - Where Null Hypothesis is True
$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \mu$$

Testing Exclusion Restrictions: The F-Test

- $$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k - 1)} \sim \mathbf{F}_{q, n-k-1}$$

where F is the test statistics (F-Statistics)

SSR is Residual Sum of Squares,

q is the number of restrictions (Numerator Degree of Freedom
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- The F-statistic measures the relative increase in SSR from the unrestricted to the restricted model.
- We will look at whether the increase in SSR in going from unrestricted model to restricted model is large enough to get the rejection.
- we will reject the H_0 if $F > c$. If H_0 is rejected then we say x_3, x_4, x_5 are jointly statistically significant at a certain significance level. If we cannot reject the null hypothesis, the variables are jointly insignificant and in most of the cases we drop them.

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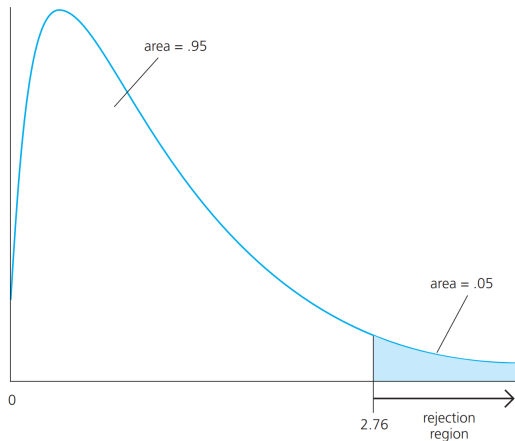
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- Note, the signs are reversed now as there is an inverse relationship between SSR and R-Squared.
- Also, Note that the F-Statistics can never be negative as $R_{ur}^2 \geq R_r^2$. However, we need to see how much the R-squared go down after the restrictions.
- Under H_0 , the numerator and denominator are independent chi-square random variables, each divided by its degrees of freedom, and such a ratio follows an F-distribution.
- Lastly, It can be shown that the F-distribution with one restriction is equivalent to the square of relevant t-distribution $t_{n-k-1}^2 = F_{1, n-k-1}$

Testing Exclusion Restrictions: The F-Test

Shape of F-Distribution

Figure 4.7 The 5% critical value and rejection region in an $F_{3,60}$ distribution.



Example

- Un-Restricted Model

$$\begin{aligned}\widehat{\log(\text{salary})} &= 11.19 + .0689 \text{ years} + .0126 \text{ gamesyr} \\ &\quad (0.29) \quad (.0121) \quad \quad (.0026) \\ &\quad + .00098 \text{ bavg} + .0144 \text{ hrunsyr} + .0108 \text{ rbisyr} \\ &\quad \quad (.00110) \quad \quad (.0161) \quad \quad (.0072) \\ n &= 353, \text{ SSR} = 183.186, R^2 = .6278,\end{aligned}$$

[4.31]

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$$\begin{aligned}\widehat{\log(\text{salary})} &= 11.22 + .0713 \text{ years} + .0202 \text{ gamesyr} \\ &\quad (.11) \quad (.0125) \quad \quad (.0013) \\ n &= 353, \text{SSR} = 198.311, R^2 = .5971.\end{aligned}\quad [4.33]$$

Example

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$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k - 1)} = \frac{(198.311 - 183.186)/3}{183.186/347} = 9.55 \sim F_{q, n-k-1}$$

Example

Un-Restricted Model

$$\begin{aligned}\widehat{\log(\text{salary})} &= 11.19 + .0689 \text{ years} + .0126 \text{ gamesyr} \\ &\quad (.029) \quad (.0121) \quad (.0026) \\ &\quad + .00098 \text{ bavg} + .0144 \text{ hrunsyr} + .0108 \text{ rbisyr} \\ &\quad (.00110) \quad (.0161) \quad (.0072) \\ n &= 353, \text{SSR} = 183.186, R^2 = .6278,\end{aligned}\quad [4.31]$$

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$$F = \frac{(\text{SSR}_r - \text{SSR}_{ur})/q}{\text{SSR}_{ur}/(n - k - 1)} = \frac{(198.311 - 183.186)/3}{183.186/347} = 9.55 \sim \mathbf{F}_{q, n-k-1}$$

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Example

- Un-Restricted Model

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- The 5% critical value of $F_{3,347}$ is about 2.63 (3.84 at 1%). As $F > c$, we reject the null hypothesis and conclude that the performance indicators are jointly significant.

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$$\begin{aligned}\widehat{\log(\text{price})} = & .264 + 1.043 \log(\text{assess}) + .0074 \log(\text{lotsize}) \\ & (.570) \quad (.151) \quad (.0386) \\ & - .1032 \log(\text{sqrft}) + .0338 \text{bdrms} \\ & (.1384) \quad (.0221) \\ n = 88, SSR = 1.822, R^2 = .773.\end{aligned}$$

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- $$F = \frac{(R^2)/k}{(1 - R^2)/(n - k - 1)} = \frac{0.773/4}{(1 - 0.773)/83} = 70.66 > c$$

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- The R-squared form cannot be used here, because the dependent variable in the restricted model $(y - x_1)$ differs from the one in the unrestricted model (y) .

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- **Rejection Rule:** $P\text{-value} < \alpha$